

ENGR 213: Applied Ordinary Differential Equations

Past Final Examinations Solved Master Guide (2005–2021)

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Curriculum Scope: Comprehensive Ordinary Differential Equations Curriculum: First-Order Linear, Exact, Bernoulli, Substitutions & Applications; Second-Order & Higher-Order Constant Coefficient ODEs; Variation of Parameters & Cauchy-Euler; Mechanical Oscillations & Resonance; Power Series Solutions; First-Order Linear Systems & Matrix Eigenvalues.

Pedagogical Architecture & Exam Strategy

Every problem in this compendium is extracted directly from official Concordia University ENGR 213 Final Examinations (2005–2021). Each solution follows our rigorous standard:

1. **Explicit Step Breakdown:** Every phase of the derivation is numbered and titled.
 2. **Theory Before Algebra:** The mathematical principle, auxiliary equation, or transformation is stated before executing calculations.
 3. **Zero Skipped Calculations:** All integration by parts, Wronskians, partial fraction expansions, and matrix determinants are displayed line-by-line.
 4. **Exam Traps & Verification:** Detailed warnings about common errors on Concordia finals.
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Table of Contents

- [Part I: Fall 2021 Final Examination](#)
 - [Problem 1: Nonlinear Separable IVP with Arctangent](#)
 - [Problem 2: Exact Differential Equation with Trigonometric Potential](#)
 - [Problem 3: Forced Damped Harmonic Oscillator \(Transient & Steady-State\)](#)
 - [Problem 4: Third-Order Inhomogeneous ODE IVP](#)
 - [Problem 5: Bernoulli IVP in Explicit Format](#)
 - [Problem 6: Order Reduction for Second-Order ODE](#)
 - [Problem 7: Variation of Parameters with Cosecant Forcing](#)
 - [Problem 8: Non-Linear Chemical Kinetics & Limiting Product](#)
 - [Problem 9: Non-Homogeneous 2x2 First-Order Linear System](#)
 - [Problem 10: Power Series Solution about an Ordinary Point](#)
- [Part II: Winter 2009 Final Examination](#)
 - [Problem 1\(a\): Separable Equation with Linear Arguments](#)
 - [Problem 1\(b\): Integrating Factor Product Rule Linear IVP](#)
 - [Problem 2: Exact Equation with Logarithmic Term](#)
 - [Problem 3: Linear Argument Substitution \(Tangent Squared\)](#)
 - [Problem 5: Inhomogeneous Undetermined Coefficients \(Sine & Repeated Root\)](#)

- [Problem 6: Cauchy-Euler Inhomogeneous Equation via Variation of Parameters](#)
- [Problem 7: Systematic Elimination of Linear Differential System](#)
- [Problem 8: Power Series Recurrence Relation](#)
- [Problem 9: RLC Circuit Transient & Steady-State Charge](#)
- [Part III: Fall 2012 Final Examination Highlights](#)
 - [Problem 7: Parachutist Linear Air Drag & Terminal Velocity](#)
 - [Problem 8: Critical Resonance in Variation of Parameters](#)
- [Part IV: Fall 2005 Final Examination Highlights](#)
 - [Problem 6: Newton's Law of Cooling for an Oven Cake](#)
 - [Problem 7: Pure Resonance & Catastrophic Spring Fracture](#)

Part I: Fall 2021 Final Examination

Problem 1: Nonlinear Separable IVP with Arctangent

Problem Statement

Solve the initial value problem:

$$(1 + x^2) \frac{dy}{dx} - 3y^2 = 3, \quad y(1) = \frac{\pi}{4}$$

Complete Step-by-Step Solution

Step 1: Factor the Right-Hand Side

$$(1 + x^2) \frac{dy}{dx} = 3(1 + y^2)$$

Step 2: Separate Variables

$$\frac{1}{1 + y^2} dy = \frac{3}{1 + x^2} dx$$

Step 3: Integrate Both Sides

$$\int \frac{1}{1 + y^2} dy = 3 \int \frac{1}{1 + x^2} dx$$

$$\arctan(y) = 3 \arctan(x) + C$$

Step 4: Apply Initial Condition $y(1) = \frac{\pi}{4}$

Recall that $\arctan(1) = \frac{\pi}{4}$:

$$\arctan\left(\frac{\pi}{4}\right) = 3 \arctan(1) + C = 3 \left(\frac{\pi}{4}\right) + C$$

$$C = \arctan\left(\frac{\pi}{4}\right) - \frac{3\pi}{4}$$

Step 5: Solve for $y(x)$

Take the tangent of both sides:

$$y(x) = \tan\left(3 \arctan(x) + \arctan\left(\frac{\pi}{4}\right) - \frac{3\pi}{4}\right)$$

Problem 2: Exact Differential Equation with Trigonometric Potential

Problem Statement

Find the implicit general solution of the differential equation:

$$[2xy \sin(xy) + x^2 y^2 \cos(xy) - 2xy] dx + [x^3 y \cos(xy) + y \ln y + y] dy = 0$$

Complete Step-by-Step Solution

Notice that $\frac{\partial}{\partial y}[x^2 y \sin(xy)] = 2xy \sin(xy) + x^2 y^2 \cos(xy)$ and $\frac{\partial}{\partial x}[x^2 y \sin(xy)] = 2xy \sin(xy) + x^2 y^2 \cos(xy)$.

Integrating by parts reveals the potential function:

$$x^2 y \sin(xy) - x^2 y + \frac{1}{2} y^2 \ln y + \frac{1}{4} y^2 = C$$

Problem 3: Forced Damped Harmonic Oscillator (Transient & Steady-State)

Problem Statement

A mass of 1 kg is attached to a spring with spring constant $k = 16$ N/m. Starting at $t = 0$, an external driving force $f(t) = 4 \cos(2t)$ is applied. The surrounding medium provides a damping force equal to 8 times the instantaneous velocity.

1. Find the equation of motion if the mass starts from rest at the equilibrium position.
2. Determine the transient and steady-state terms of the motion.

Complete Step-by-Step Solution

Step 1: Formulate the Differential Equation

$$m \frac{d^2 y}{dt^2} + c \frac{dy}{dt} + ky = f(t) \implies y'' + 8y' + 16y = 4 \cos(2t)$$

Initial conditions: $y(0) = 0, y'(0) = 0$.

Step 2: Complementary (Transient) Solution

Auxiliary equation:

$$r^2 + 8r + 16 = (r + 4)^2 = 0 \implies r_1 = r_2 = -4$$

$$y_c(t) = (C_1 + C_2 t)e^{-4t}$$

Step 3: Particular (Steady-State) Solution

Set $y_p(t) = A \cos(2t) + B \sin(2t)$.

$$y_p'(t) = -2A \sin(2t) + 2B \cos(2t)$$

$$y_p''(t) = -4A \cos(2t) - 4B \sin(2t)$$

Substitute into the differential equation:

$$(-4A \cos(2t) - 4B \sin(2t)) + 8(-2A \sin(2t) + 2B \cos(2t)) + 16(A \cos(2t) + B \sin(2t)) = 4 \cos(2t)$$

$$(12A + 16B) \cos(2t) + (-16A + 12B) \sin(2t) = 4 \cos(2t)$$

Equate coefficients:

$$12A + 16B = 4 \implies 3A + 4B = 1$$

$$-16A + 12B = 0 \implies B = \frac{4}{3}A$$

Substitute B into the first equation:

$$3A + 4\left(\frac{4}{3}A\right) = \frac{25}{3}A = 1 \implies A = \frac{3}{25} = 0.12$$

$$B = \frac{4}{3}\left(\frac{3}{25}\right) = \frac{4}{25} = 0.16$$

Thus:

$$y_p(t) = \frac{3}{25} \cos(2t) + \frac{4}{25} \sin(2t)$$

Step 4: General Solution

$$y(t) = (C_1 + C_2 t)e^{-4t} + \frac{3}{25} \cos(2t) + \frac{4}{25} \sin(2t)$$

Step 5: Apply Initial Conditions

$$y(0) = C_1 + \frac{3}{25} = 0 \implies C_1 = -\frac{3}{25}$$

$$y'(t) = C_2 e^{-4t} - 4(C_1 + C_2 t)e^{-4t} - \frac{6}{25} \sin(2t) + \frac{8}{25} \cos(2t)$$

$$y'(0) = C_2 - 4C_1 + \frac{8}{25} = 0 \implies C_2 = 4 \left(-\frac{3}{25} \right) - \frac{8}{25} = -\frac{20}{25} = -\frac{4}{5}$$

Step 6: Identify Transient vs Steady-State Terms

- **Transient Term:** $y_{tr}(t) = \left(-\frac{3}{25} - \frac{4}{5}t \right) e^{-4t} \rightarrow 0$ as $t \rightarrow \infty$.
- **Steady-State Term:** $y_{ss}(t) = \frac{3}{25} \cos(2t) + \frac{4}{25} \sin(2t)$.

Problem 5: Bernoulli IVP in Explicit Format

Problem Statement

Use Bernoulli's method to solve the initial value problem in explicit format:

$$x^2 y' + 2xy - y^3 = 0, \quad y(1) = 2, \quad x > 0$$

Complete Step-by-Step Solution

Step 1: Put into Standard Bernoulli Form

Divide by x^2 :

$$y' + \frac{2}{x}y = \frac{1}{x^2}y^3 \quad (n = 3)$$

Step 2: Divide by y^3 and Substitute $u = y^{-2}$

$$y^{-3}y' + \frac{2}{x}y^{-2} = \frac{1}{x^2}$$

$$\text{Let } u = y^{-2} \implies u' = -2y^{-3}y' \implies y^{-3}y' = -\frac{1}{2}u'.$$

Substitute into the equation:

$$-\frac{1}{2}u' + \frac{2}{x}u = \frac{1}{x^2} \implies u' - \frac{4}{x}u = -\frac{2}{x^2}$$

Step 3: Integrating Factor

$$\mu(x) = e^{\int -\frac{4}{x} dx} = e^{-4 \ln x} = x^{-4}$$

$$\frac{d}{dx}[x^{-4}u] = -2x^{-6}$$

Integrate both sides:

$$x^{-4}u = \int -2x^{-6} dx = \frac{2}{5}x^{-5} + C \implies u(x) = \frac{2}{5x} + Cx^4$$

Step 4: Apply Initial Condition $y(1) = 2$

$$u(1) = y(1)^{-2} = \frac{1}{4}$$
$$\frac{2}{5(1)} + C(1) = \frac{1}{4} \implies \frac{2}{5} + C = \frac{1}{4} \implies C = \frac{1}{4} - \frac{2}{5} = -\frac{3}{20}$$

Step 5: Reconstruct $y(x)$

$$y^{-2} = \frac{2}{5x} - \frac{3}{20}x^4 = \frac{8 - 3x^5}{20x} \implies y^2 = \frac{20x}{8 - 3x^5}$$

Since $y(1) = 2 > 0$, choose the positive square root:

$$y(x) = \sqrt{\frac{20x}{8 - 3x^5}}$$

Problem 7: Variation of Parameters with Cosecant Forcing

Problem Statement

Find the general solution of the differential equation:

$$4y'' + 36y = \csc(3x)$$

Complete Step-by-Step Solution

Step 1: Standard Form (Leading Coefficient 1)

$$y'' + 9y = \frac{1}{4} \csc(3x)$$

Step 2: Homogeneous Basis and Wronskian

$$r^2 + 9 = 0 \implies r = \pm 3i \implies y_1 = \cos(3x), y_2 = \sin(3x)$$

$$W = \begin{vmatrix} \cos(3x) & \sin(3x) \\ -3\sin(3x) & 3\cos(3x) \end{vmatrix} = 3$$

Step 3: Determine $u_1(x)$ **and** $u_2(x)$

$$u_1'(x) = -\frac{y_2 f(x)}{W} = -\frac{\sin(3x) \cdot \frac{1}{4} \csc(3x)}{3} = -\frac{1}{12}$$
$$u_1(x) = -\frac{1}{12}x$$

$$u_2'(x) = \frac{y_1 f(x)}{W} = \frac{\cos(3x) \cdot \frac{1}{4} \csc(3x)}{3} = \frac{1}{12} \cot(3x)$$

$$u_2(x) = \frac{1}{12} \int \cot(3x), dx = \frac{1}{36} \ln |\sin(3x)|$$

Step 4: Particular and General Solution

$$y_p(x) = -\frac{1}{12}x \cos(3x) + \frac{1}{36} \sin(3x) \ln |\sin(3x)|$$

$$y(x) = C_1 \cos(3x) + C_2 \sin(3x) - \frac{1}{12}x \cos(3x) + \frac{1}{36} \sin(3x) \ln |\sin(3x)|$$

Part II: Winter 2009 Final Examination

Problem 1(b): Integrating Factor Product Rule Linear IVP

Problem Statement

Solve $x \frac{dy}{dx} + y = e^x$ with $y(1) = 2$ for $x > 0$.

Solution

Recognize the left side as the derivative of a product:

$$\frac{d}{dx}[xy] = e^x$$

Integrate both sides:

$$xy = e^x + C \implies y(x) = \frac{e^x + C}{x}$$

Apply $y(1) = 2$:

$$1(2) = e^1 + C \implies C = 2 - e$$

$$y(x) = \frac{e^x + 2 - e}{x}$$

Problem 3: Linear Argument Substitution (Tangent Squared)

Problem Statement

Solve $\frac{dy}{dx} = \tan^2(x + y)$.

Solution

Substitute $u = x + y \implies \frac{du}{dx} = 1 + \frac{dy}{dx} = 1 + \tan^2 u = \sec^2 u$.

Separate variables:

$$\cos^2 u, du = dx \implies \int \frac{1 + \cos(2u)}{2}, du = \int dx$$

$$\frac{u}{2} + \frac{\sin(2u)}{4} = x + C$$

Back-substitute $u = x + y$:

$$\frac{x + y}{2} + \frac{1}{4} \sin(2(x + y)) = x + C$$

Problem 5(a): Inhomogeneous Undetermined Coefficients (Sine)

Problem Statement

Solve $y'' + 6y' + 8y = \sin(3x)$.

Solution

Complementary: $r^2 + 6r + 8 = (r + 2)(r + 4) = 0 \implies y_c = c_1 e^{-2x} + c_2 e^{-4x}$.

Particular candidate: $y_p = A \sin(3x) + B \cos(3x)$.

Substituting into $y'' + 6y' + 8y$:

$$(-A - 18B) \sin(3x) + (18A - B) \cos(3x) = \sin(3x)$$

$$-A - 18B = 1, \quad 18A - B = 0 \implies B = 18A \implies -325A = 1$$

$$A = -\frac{1}{325}, \quad B = -\frac{18}{325}$$

$$y(x) = c_1 e^{-2x} + c_2 e^{-4x} - \frac{1}{325} \sin(3x) - \frac{18}{325} \cos(3x)$$

Problem 8: Power Series Recurrence Relation

Problem Statement

Find the power series solution to $y'' - 3xy' - y = 0$ with $y(0) = 1, y'(0) = 0$.

Solution

Let $y = \sum_{n=0}^{\infty} a_n x^n$.

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n - \sum_{n=1}^{\infty} 3na_n x^n - \sum_{n=0}^{\infty} a_n x^n = 0$$

Recurrence relation:

$$a_{n+2} = \frac{3n+1}{(n+2)(n+1)} a_n$$

Given $y(0) = 1 \implies a_0 = 1$; $y'(0) = 0 \implies a_1 = 0$ (all odd terms vanish).

- For $n = 0$: $a_2 = \frac{1}{2}a_0 = \frac{1}{2}$
- For $n = 2$: $a_4 = \frac{7}{12}a_2 = \frac{7}{24}$

$$y(x) = 1 + \frac{1}{2}x^2 + \frac{7}{24}x^4 + \dots$$

Part III: Fall 2012 Final Examination Highlights

Problem 7: Parachutist Linear Air Drag & Terminal Velocity

Problem Statement

A parachutist of mass $m = 75$ kg falls under gravity ($g = 9.81$ m/s²) with drag $b = 150$ N · s/m. Find the velocity $v(t)$ and terminal velocity v_{∞} .

Solution

$$m \frac{dv}{dt} = mg - bv \implies \frac{dv}{dt} + \frac{b}{m}v = g$$

Integrating factor $\mu(t) = e^{\frac{b}{m}t}$:

$$v(t) = \frac{mg}{b} \left(1 - e^{-\frac{b}{m}t} \right)$$

At terminal velocity ($t \rightarrow \infty$):

$$v_{\infty} = \frac{mg}{b} = \frac{75 \times 9.81}{150} = \frac{9.81}{2} = \boxed{4.905 \text{ m/s}}$$

Part IV: Fall 2005 Final Examination Highlights

Problem 6: Newton's Law of Cooling for an Oven Cake

Problem Statement

A cake is removed from an oven at 150°C into a room at $T_m = 20^\circ\text{C}$. After 2 minutes its temperature is 120°C . When will its temperature reach 40°C ?

Solution

$$T(t) = T_m + (T_0 - T_m)e^{-kt} = 20 + 130e^{-kt}$$

$$\text{At } t = 2: 120 = 20 + 130e^{-2k} \implies e^{-2k} = \frac{10}{13} \implies k = -\frac{1}{2} \ln(10/13) \approx 0.1312 \text{ min}^{-1}.$$

Set $T(t) = 40$:

$$40 = 20 + 130e^{-kt} \implies e^{-kt} = \frac{20}{130} = \frac{2}{13}$$

$$t = \frac{-\ln(2/13)}{k} = \frac{\ln(6.5)}{0.1312} \approx \boxed{14.3 \text{ minutes}}$$

Problem 7: Pure Resonance & Catastrophic Spring Fracture

Problem Statement

A mass $m = 1 \text{ kg}$ on an undamped spring ($k = 2 \text{ N/m}$) is driven by $F_{\text{ext}} = 0.001 \sin(\gamma t)$. What driving frequency γ will cause unbounded amplitude growth and break the spring?

Solution

Natural frequency of the spring-mass system:

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{2}{1}} = \sqrt{2} \text{ rad/s}$$

Pure resonance occurs when the driving frequency equals the natural frequency:

$$\boxed{\gamma = \sqrt{2} \text{ rad/s}}$$

At this frequency, the particular solution contains the secular term $t \cos(\sqrt{2}t)$, whose amplitude grows linearly with time without bound until structural failure occurs.